

Gustavo Pinho Maia

PhD Researcher in Chemistry · Astrobiology

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CiênciaVitae 8212-CEE3-E928

Education

2023–2027 **PhD in Chemistry (Astrobiology)**
Ongoing · Universidade de Lisboa · Instituto Superior Técnico
Mechanochemical energy and prebiotic synthesis

2020–2022 **MSc in Chemistry**
Concluded · Instituto Superior Técnico
Ribonucleoside stability and prebiotic synthesis

2017–2020 **BSc in Industrial Chemistry**
Universidade da Beira Interior
Optimization of quinoline N-oxide synthesis

2013–2016 **Specialization Level IV as a Mechatronics Technician**
Concluded · Escola Profissional de Espinho
Fully automated robot for Rescue Line Competition (RoboCupJunior)

Research & professional experience

Jan 2025 – Present **Visiting Scientist**
MNHN · IMPMC, France

Nov 2024 – Dec 2024 **Visiting Scientist**
NASA Goddard Space Flight Center, USA

Sep 2024 – Oct 2024 **Invited Assistant (University Teacher)**
Instituto Superior Técnico, Portugal

Jun 2021 – May 2022 **President of the Fiscal Council**
ANEQ, Portugal

Jun 2020 – Jun 2021 **Vice President**
ANEQ, Portugal

Sep 2019 – Dec 2020 **Vice-President, UBIQUÍMICA**
University of Beira Interior

Jul 2020 – Sep 2020 **Research Intern**
Universidade da Beira Interior

Oct 2019 – Sep 2020 **Delegate of Industrial Chemistry Students**
Universidade da Beira Interior

Aug 2019 – Sep 2019 **Factory Worker**
Nestlé Portugal

Aug 2018 – Aug 2019 **Scientific Committee Member, UBIQUÍMICA**
University of Beira Interior

Jun 2014 – Jul 2014 **Professional Internship**
Ramada Aços, S.A.

Jun 2016 – Jul 2016 **Participant, RoboCup 2016 World Championship (Junior Rescue League)**
World robotics championship · Leipzig, Germany

May 2016 **Participant, XVI RoboCup Portugal Open**
National robotics competition · Bragança, Portugal

2015 **Participant, XV RoboCup Portugal Open**
National robotics competition · Vila Real, Portugal

May 2014 **Participant, XIV RoboCup Portugal Open**
National robotics competition · Espinho, Portugal

Publications

1. **Gustavo P. Maia** et al. (2026). **Effect of asteroid gardening on the exogenous delivery of extraterrestrial organic matter**. *Earth and Planetary Science Letters*. doi:10.1016/j.epsl.2026.120237
2. **Gustavo P. Maia**, Catarina Gonçalves, Ana J. Carvalho, Vânia André, Adelino Galvão, Ana P. C. Ribeiro, Pedro F. Pinheiro and José Armando Luísa da Silva (2025). **Mechanochemical Reactivity of Ribonucleosides Mediated by Inorganic Species: Implications for Extraterrestrial Organic Matter Interpretation**. *Applied Sciences*. doi:10.3390/app15031363
3. **Gustavo P. Maia**, José Armando Luísa da Silva, Vânia André and Adelino M. Galvão (2023). **Shock-Induced Degradation of Guanosine and Uridine Promoted by Nickel and Carbonate: Potential Applications**. *Molecules*. doi:10.3390/molecules28248006
4. Ana Franco, **Gustavo P. Maia** and José Armando Luísa da Silva (2023). **Da Química Bioinorgânica para a Química Prebiótica: Levar Mais Longe o Legado do Prof. Fraústo da Silva**. *Boletim da Sociedade Portuguesa de Química*. doi:10.52590/M3.P708.A30002745

Presentations

Aug 2026 **Effect of Mechanochemistry on Organic Matter: Implications for Impact Gardening on Asteroid Surfaces** ORAL
MetSoc 2026 · Frankfurt
Oral talk · Maia, Remusat, Amano, Viennet, Dworkin, McLain & da Silva

Oct 2025 **The effect of mechanochemistry on the formation of extraterrestrial organic matter** ORAL
AbGradE'25 · Lisbon
Oral talk · Maia, Dworkin, da Silva & Remusat

2025 **Solid-state mechanochemistry of extraterrestrial organic molecules: new insights on the exogenous delivery of organic matter** POSTER
EANA 2025 · Lisbon · Poster Award
Award-winning poster · EANA 2025

Jul 2025

Effect of mechanochemical events on extraterrestrial organic matter and its implications for exogenous delivery ORAL

BEACON 2025 · Harpa, Reykjavik

Oral presentation · BEACON 2025

2025

Mechanochemistry of small bodies and the evolution of extraterrestrial organic matter ORAL

IPGP "Small Bodies Day" 2025 · Paris

Oral talk · IPGP, Paris

23 Jun 2025

The effect of shock/impacts on the exogenous delivery of organic matter ORAL

IMPMC PhD Day · Paris

Oral talk · IMPMC, Sorbonne Université

28 Jan 2025

Mechanochemistry of extraterrestrial bodies: Bridging exogenous delivery with prebiotic chemistry ORAL

MNHN · Paris

Oral talk · MNHN, Paris

28 Jan 2025

Mechanochemistry and astrobiology: mimicking impact reactions ORAL

MNHN · Paris

Oral talk · MNHN, Paris

2024

Impact events of extraterrestrial bodies: the bridge between exogenous delivery and the origin of life ORAL

NInTec Science Days 2024 · Lisbon

Oral talk · NInTec Science Days 2024

2024

Shock-Impact Events and the Origin of Life: a mechanochemical approach POSTER

IST PhD Open Days 2024 · Lisbon

Poster · PhD Open Days 2024

Aug–Sep 2022

Ribonucleoside stability and prebiotic synthesis ORAL

8th EuChemS Chemistry Congress · CCL, Lisbon

Oral talk · 8th EuChemS Congress

2022

Ribonucleosides stability and prebiotic synthesis POSTER

CQE Days 2022 · Lisbon · P083

Poster · CQE Days 2022

Awards

2025

EANA 2025 Poster Award

EANA 2025 · Lisbon, Portugal

2017

Diploma of Merit Recognition

Rotary Club de Espinho

2016

Best Presentation · RCJ Rescue Maze

RoboCup 2016 · Leipzig, Germany

Funding

2023

Mechanochemical energy and prebiotic synthesis

Doctoral Grant · Fundação para a Ciência e a Tecnologia (FCT) · 2023.01099.BD

Jun – Nov 2023

A Gold Nanosensor for Single Molecule Detection of RNA Biomarkers in a Miniaturized Microscope
Research Fellow · Centro de Química Estrutural, Universidade de Lisboa

2020

Centro de Química Estrutural
Research Unit Funding · Fundação para a Ciência e a Tecnologia (FCT) · UIDB/00100/2020

Courses & training

22–26 June 2026

Building the Hard Rocky Planets — From Mercury to the Asteroid Belt
Summer School · Institut Origines (Aix-Marseille Université) & Institut de Physique du Globe de Paris (IPGP) · Marseille, France

June 2026

Astrobiology Introductory Course (Virtual)
Course · Rencontres Exobiologiques pour Doctorants (RED) · Le Teich, France · Online

March 2025

Astrobiology Introductory Course
Course · Rencontres Exobiologiques pour Doctorants (RED) · Le Teich, France

February 2025

Recommendations for Publishing in International Journals
Formation · Biblioteca do Instituto Superior Técnico (BIST) · Online

Techniques & software

Analytical techniques
Chromatography & MS, NMR, XRD, FTIR, EA, irMS, SEM · EDX

Synthesis & modelling
Mechanochemistry, Organic Synthesis, Ab Initio

Engineering & electronics
Arduino, PICAXE, Multisim, SOLIDWORKS, AutoCAD, SketchUp

Instruments & data
TopSpin, MestReNova, OpenLab, MassLynx & TargetLynx, OPUS, Profex, ACD/Labs

Chemistry & computation
Python, R, MATLAB, ChemDraw, Chemcraft

Productivity & code
Office 365, VS Code, GitHub